

DC-AC Pure Sine Wave Inverters

Clean, Stable AC Power from Rugged DC Sources

Stable Pure Sine Output.
High-Power AC Conversion.
Built for Demanding Applications.



CRYONANO DC-AC Pure Sine Wave Inverters are engineered to deliver clean, stable, and grid-quality AC power from onboard DC sources in mission-critical environments.

Designed for railways, defense vehicles, aerospace platforms, marine systems, and industrial infrastructure, these inverters convert battery and DC bus power into low-distortion AC output suitable for sensitive electronics, communication systems, HVAC equipment, control units, and operational loads.

Built with rugged industrial architecture and optimized thermal performance, the systems ensure reliable operation under harsh environmental and electrical conditions.

Notes

- Charging performance depends on battery chemistry and configuration
- Custom charging profiles available for specialized applications
- Designed for continuous operation in harsh environments
- Thermal management varies by power level and enclosure type

KEY FEATURES

• Power & Performance

Delivers 30 kW / 45 kW / up to 60 kW output power with 10–600 V output range, up to 1000 A maximum current, compact rack form factor, and low-noise operation for sensitive environments.

• Advanced Control

Supports programmable voltage and current slew rates, CPL operation, pre-load enable/disable, dedicated control pins, output disconnect, and polarity reversal.

• User Interface & Connectivity

Multi-function front panel with LAN, USB 2.0, RS-232 / RS-485, and isolated analog interface (5V / 10V) enables flexible system integration.

• Optional Interfaces

Optional IEEE (GPIB), EtherCAT, Modbus-TCP, and isolated 4–20 mA (IS420) interfaces provide advanced communication and automation capabilities.

APPLICATIONS

- Locomotives and railway systems
- Metro and passenger rail platforms
- Defense and military vehicles
- Marine electrical systems
- Signaling and communication infrastructure
- Industrial mobile power systems
- Aerospace onboard support systems



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SPECIFICATIONS

ELECTRICAL PERFORMANCE

Output Power Up to 10 kW

Input Type Onboard DC supply

Output Type Pure sine wave AC

Output Stability Low-distortion regulated output

Conversion Type DC to AC inversion

SYSTEM PERFORMANCE

EMI Performance Low electromagnetic interference

Thermal Management High-efficiency cooling

Load Compatibility Sensitive and mixed AC loads

Operational Mode Continuous-duty capable

MECHANICAL & ENVIRONMENTAL

Construction Rugged industrial-grade

Cooling Method Configuration dependent

Integration Embedded / rack / vehicle systems

Environment Transportation, marine, defense-grade

CONFIGURABILITY

Architecture Options MOTS / Modified / Fully Custom

Voltage Configuration Application dependent

Mounting Rack / frame / embedded

Customization Cooling, packaging, environmental options

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